Engine Model(s): PT6A

72-30-01

CENTER FIRESEAL MOUNT RING - MAINTENANCE PRACTICE

1. General

A. It is not a necessary practice to remove the mount ring during maintenance of the engine. Removal should be dictated by the extent of repair necessary as determined during inspection. Refer to the relevant chapters in this manual for the removal/installation of lines/tubes and components passing through the rings.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-101	Cloth, Abrasive
PWC08-002	Adhesive, Fluorosilicone
PWC08-002A	Adhesive, Fluorosilicone
PWC11-002	(Use PWC11-014)
PWC11-014	Alcohol, Isopropyl

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation

A. Removal of Pre-SB1445 Center Fireseal Mount Ring (Ref. Fig. 202)

(1) Remove external lines and components passing through fireseal mount ring (Ref. Relevant Chapters).

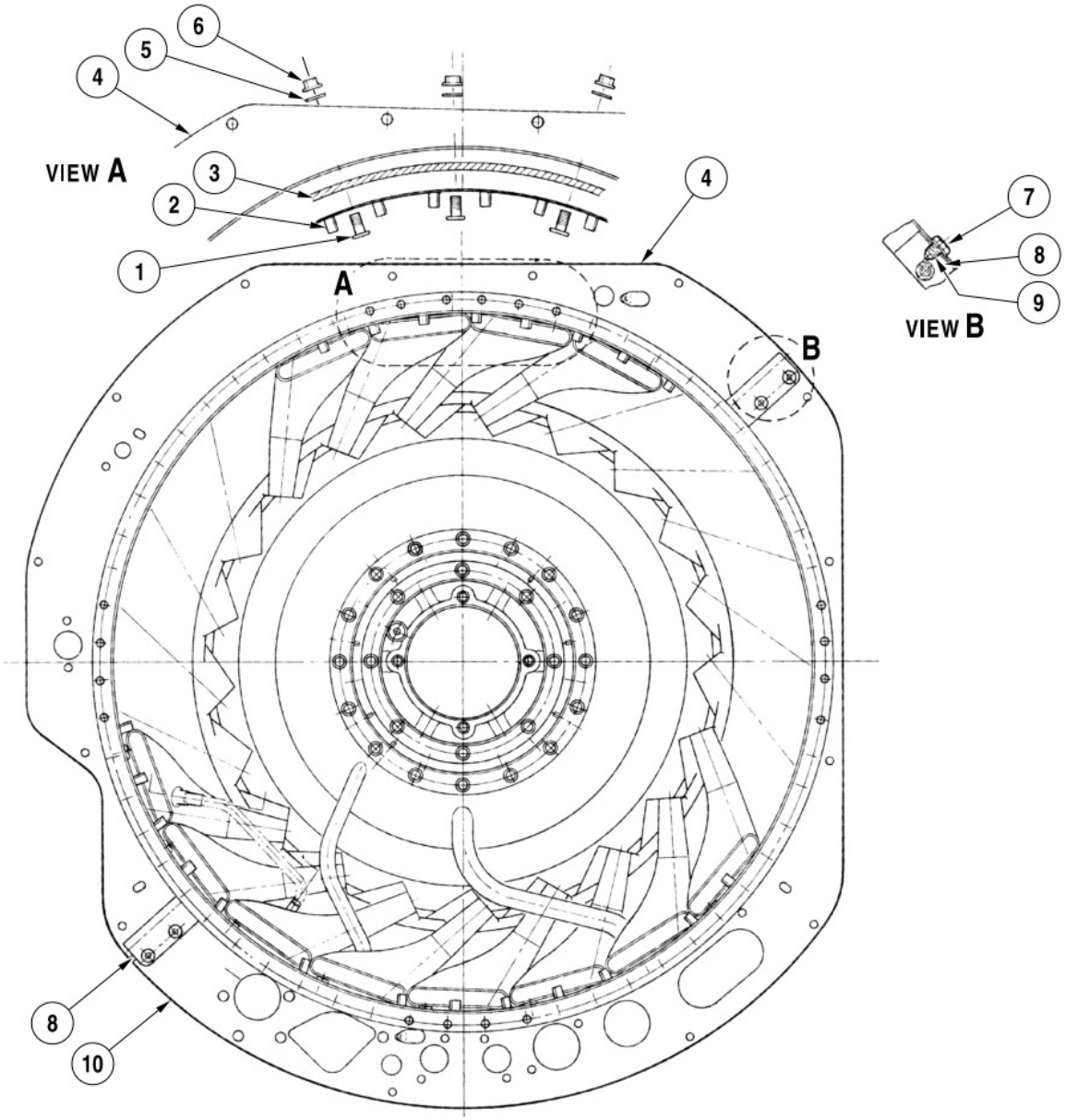
(2) Remove nuts and bolts joining ring halves at flanges.

(3) Remove nuts, washers and bolts that secure ring halves to brackets on gas generator case. Remove ring halves.

B. Removal of Post-SB1445 Center Fireseal Mount Ring (Ref. Fig. 201)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 201 Removal/Installation of Center Fireseal Mount Ring Assembly



Post-SB1445

c41527

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Bolt
2. Vane Segment Strip
3. Gas Generator Case
4. Fireseal Mount Ring, Top Half
5. Washer
6. Nut, Self-locking
7. Bolt
8. Flange, Joining
9. Nut, Self-locking
10. Fireseal Mount Ring, Bottom Half

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(1) Remove the power section (Ref. 72-00-00, REMOVAL/INSTALLATION).

(2) Remove the external lines and components passing through the fireseal mount ring (Ref. Relevant Chapters).

(3) Remove the combustion chamber liner (Ref. 72-40-01, Removal/Installation).

(4) Remove the nuts (9) and bolts (7) joining the fireseal ring halves (4 and 10).

CAUTION: EXTREME CARE SHOULD BE TAKEN AT DISASSEMBLY, THAT BOLTS SECURING THE FIRESEAL RING HALVES DO NOT FALL INTO THE GAS GENERATOR CASE.

(5) Remove the nuts (6), washers (5) and bolts (1) securing the ring halves (4 and 10) and vane segment strips (2) to the gas generator case (3).

(6) Remove the vane segment strips (2) and note the location for reinstallation.

(7) Remove the fireseal ring halves (4 and 10).

NOTE: A sharp blow with a soft faced hammer may be necessary to break the adhesive bond.

C. Installation of Pre-SB1445 Center Fireseal Mount Ring (Ref. Fig. 202)

(1) Position the ring halves on the gas generator case and secure to the mounting brackets with bolts, washers and nuts. Tighten and torque the nuts 32 to 36 lb.in.

(2) Join the ring halves at the flanges with bolts and self-locking nuts. Insert the bolts from the bottom on left side and from the top on the right side. Tighten and torque the nuts 32 to 36 lb.in.

(3) Install the external lines and components passing through the fireseal mount ring (Ref. Relevant Chapters).

D. Installation of Post-SB1445 Center Fireseal Mount Ring (Ref. Fig. 201)

(1) Before the assembly, apply silicone adhesive (PWC08-002) or (PWC08-002A) to the mating surfaces surrounding the ring mounting holes in the gas generator case and to the shank of the mounting bolts.

(2) Locate the top ring half (4) on the gas generator case (3), with the mounting holes aligned with the holes in the gas generator case.

(3) Locate the vane segment strip (2) inside the gas generator case (3) in the position noted on removal. With the mounting holes aligned, secure with bolts (1) inserted from the inside. Install and tighten the washers (5) and nuts (6) fingertight.

(4) Locate the bottom ring half (10) on the gas generator case (3) with the mounting holes aligned with the holes in the gas generator case. Install the vane segment strip (2) (Ref. Step (3)).

(5) Secure the top and bottom ring halves (4 and 10) at the joining flanges (8) with the bolts (7) and nuts (9) tightened fingertight.

NOTE: Dependent on the aircraft installation, two outboard bolts, one on each fireseal lap joint, may be removed to overcome interference with the airframe fireseal.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(6) Install the remaining vane segments strips (2) (Ref. Step (3)) and secure with the bolts (1) inserted from the inside. Tighten the washers (5) and nuts (6) and torque 45 to 55 lb.in. for all twenty-one locations.

(7) Tighten the joining flange nuts (9), as applicable, 32 to 36 lb.in.

(8) Remove any excessive silicone adhesive.

(9) Install the combustion chamber liner (Ref. 72-40-01, Removal/Installation).

(10) Install the power section (Ref. 72-00-00, REMOVAL/INSTALLATION).

(11) Install the external lines and components passing through the fireseal mount ring (Ref. Relevant Chapters).

6. Inspection/Check

A. Center Fireseal Mount Ring

(1) Examine the mount ring halves (Pre-SB1445 or Post-SB1445) for attachment, damage and condition.

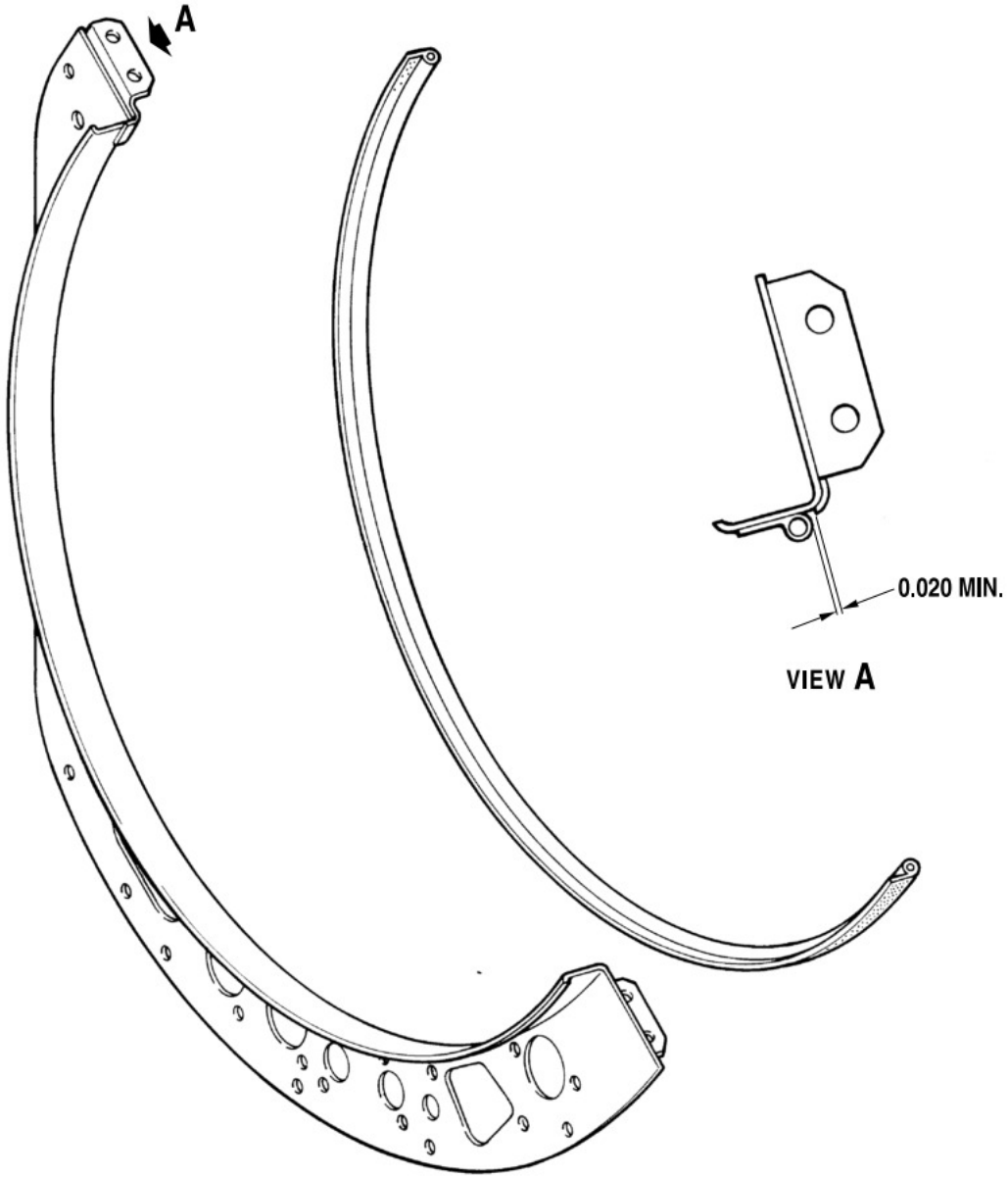
NOTE: For the external tubes/lines passing through the mount rings, refer to relevant chapters in this manual.

(2) Examine the circumferential packing material for security (Pre-SB1445 or Post-SB1445). A loose strip may be rebonded (Ref. Approved Repairs).

7. Approved Repairs (Pre-SB1445) (Ref. Fig. 202)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 202 Center Fireseal Mount Ring - Insulating Strip Replacement



Pre-SB1445

c7050

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

A. Center Fireseal Mount Ring Packing Material

(1) A detached fireseal packing material, if serviceable, may be rebonded:

(a) Clean surfaces to be bonded with alcohol (PWC11-002) or (PWC11-014).

(b) Roughen the surface of the packing material with abrasive cloth (PWC05-101), and clean with alcohol (PWC11-002) or (PWC11-014).

(c) Bond the packing material to the center ring halves using adhesive sealant (PWC08-002).

(d) Allow the sealant to cure at room temperature for approximately 72 hours.

NOTE: The ring half may be returned to service after two to four hours elapsed time, but avoid exerting pressures that would cause the sealant to exude and weaken the bond.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement